ECE 510 Hardware Project

A 10-bit Synchronous Up/Down Counter

Introduction

- This project requires the use of the Basys3 FPGA development board by Digilent and is due 4-23-25 (Wednesday).
- UofL confirmation of your design must be verified using an in-person demo with Ben Eells (benjamin.eells@louisville.edu) or Eugene Rockey. The demonstration must be completed **before 5pm on Wednesday, 4-23-25**.
- Your hardware project report **must** be submitted on Blackboard by midnight on 4-23-25.
- Late reports or demonstrations will not be accepted.
- You will need to make use of the Basys3 Reference Manual for configuring the switches, buttons (de-bounced) and four 7-segment displays WITHOUT the flicker.
- In this project, you will create a counter design. Then you will interface the design with most of the I/O available on the Basys3 development board. You will develop a 10-bit counter that has the capability to count up and down from 0 to 1000 (10-bit). The design will utilize the slide switches as inputs, and it will utilize the 7-segment display and LEDs for displaying the output.

Objectives

The objectives of this project are as follows:

- To develop the understanding of the 7-segment display and how to interface with it.
- Show your understanding of VHDL design components such as counters and how to implement them into existing designs.
- Test your understanding of VHDL and your ability to apply concepts taught in class to real-world engineering applications.

Figure 1: Basys3 FPGA board.

Design Description

This project consists of a counter programmed with VHDL based on the example reviewed in class.

1. This project requires the student to implement a 10-bit counter design using VHDL. The counter will have the ability to count up and down from 0 and 1000. This requires the utilization

of 16 slide switches and the center push button of the Basys3 board. The 10 right-most switches (SW0 – SW9) will be used to input a preset value to the counter or the ALU. SW0 represents the LSB and SW9 represents the MSB of the input value.

The center push button (BTNC) is used as the manual clock for the counter. The count should increase or decrease (depending on the selector switch, SW13) by 1 on every rising edge of this button.

SW14 will be a **synchronous preset to the counter**. SW15 will be an **asynchronous reset**. If the asynchronous reset goes "high" (logic '1', switch slides to the top), the count should reset to 0 regardless of the clock signal. As long as the reset stays "high", the count should stay at 0 even if the clock is toggled or the preset is applied.

The synchronous preset should set the count value to the value on the slide switches SW0-SW9. This is synchronous, so nothing should happen to the count until the preset goes "high" (logic '1') AND the clock signal (BTNC) is toggled.

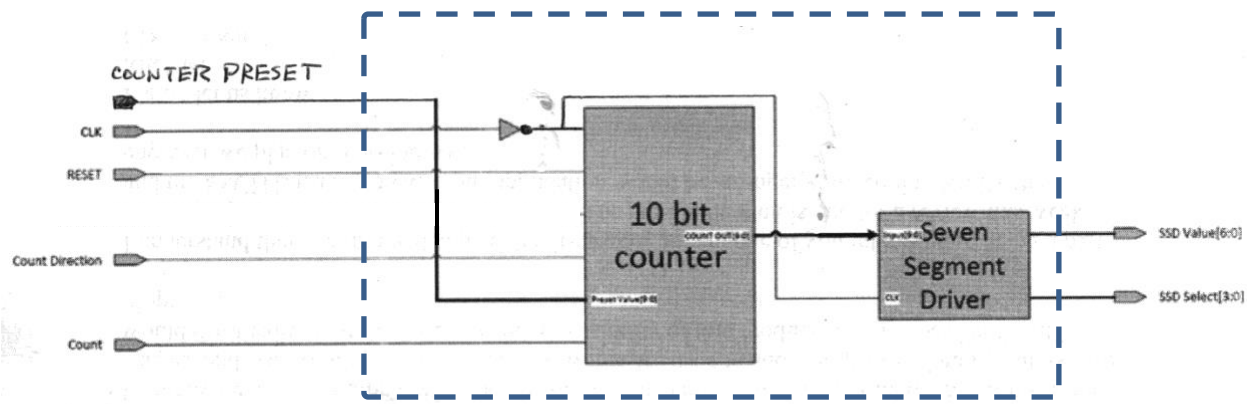


Figure 2: Simple schematic of the design, showing the components and the black-box outline.

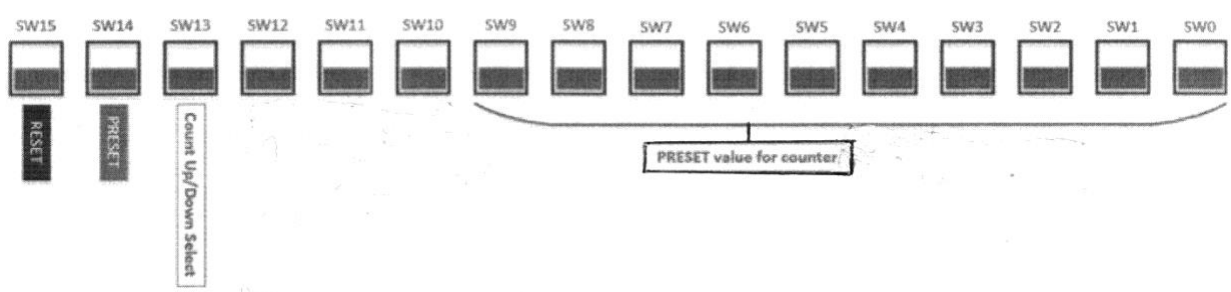


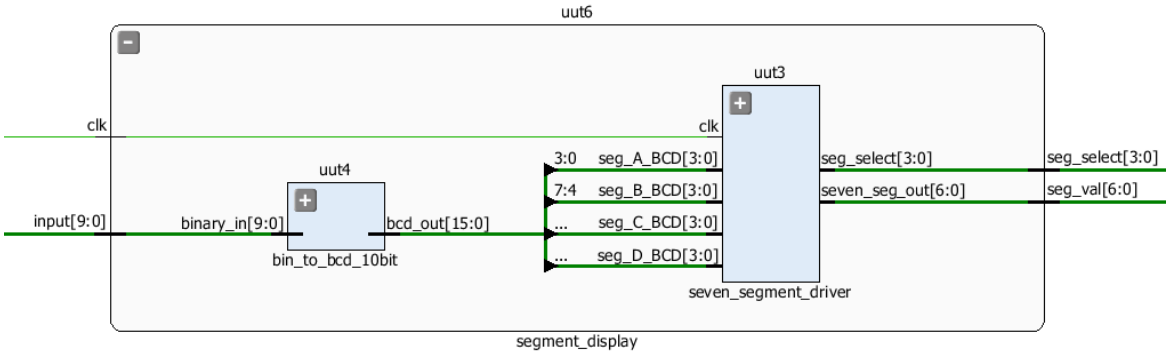
Figure 3: Slide switch map for the project

Helpful hints:

- **Read the CLK through a NOT gate** as shown in the figure 2 to avoid a possible error in synthesis.

- There is no need to de-bounce the slide switches. However, the push button (BTNC) that you use as the clock has to be de-bounced. You can do this by simply using a D-FlipFlop. For a better de-bouncing circuit, you can use a provided language template within Vivado. To do this within Vivado, go to Window → Language Templates. Then go to, VHDL→ Synthesis Constructs → Coding examples → Misc and choose Debounce Circuit.

- It would be helpful to develop a binary to BCD conversion algorithm to display the count easier. The output of the BCD conversion should be a 16-bit long vector (4 bits per digit). Then you can perform a BCD to seven segment conversion to display the BCD value as a number in the SSD. (For more information on driving the SSD read section 8.1 in the Basys3 reference manual)



- Using a hierarchical design with multiple vhd files (one for the counter process, one for the binary to BCD conversion, one for the SSD control, etc…) would be helpful to simplify your design process. These can be called using component declarations in the architecture block before the signal declarations and begin statement.

Project Sign-off Requirements

In order to receive full credit for this project assignment the design must be able to complete the following 8 tests:

1. If SW13=0 and BTNC is pushed, the SSD value must increase by 1.
2. If SW13=1 and BTNC is pushed, the SSD value must decrease by 1.
3. The SSD flicker must no longer be present.
4. The count value must count from 0 to 1000 then reset back to 0. It must not count any higher than 1000.
5. The counter reset must be asynchronous and the preset must be synchronous.
6. If the reset switch (SW15) is toggled "high", the count must reset to 0 immediately and stay at zero until the reset switch is toggled "low". Once the switch is "low", the counting process can resume as normal.
7. The preset must be dynamic. In other words, if the synchronous preset slide switch (SW14) is set to "high" and the BTNC is pushed, the output of the counter must be set to the 10-bit binary value read in from the slide switches (SW0-SW9).
8. If a preset value higher than the decimal equivalent of 1000 is entered using the slide switches (i.e. "1111111111" or 1023), count must be set to 1000 during preset function.

Report Requirements

Minimum of 5 pages not including figures or appendix, which includes your VHDL code. The report must include the following:

1. An introduction with a schematic of the VHDL design from Vivado.
2. A description of the VHDL code and modifications made to existing code from Labs 6, 7 and 8.
3. A discussion of the results and how long the project took in hours.
4. How many iterations of this design can fit in the FPGA if I/O weren't a limitation?
5. Any problems or obstacles that were encountered throughout the process and how you solved them.
6. The VHDL code must be submitted as an appendix to your report and does not count toward the 5-page requirement.